HW4 Rigid Body Simulator Testing Log

I worked with Codex to test my HW3 simulator, a translating 2D circle under gravity with floor and side-wall collisions. The model has no rotation. I used unittest and independent expected values, with the new tests stored in HW4.

Analytical solutions

I checked an upward launch through ascent, apex, and descent. I explicitly tested the semi-implicit Euler height bias, $-gTh/2$, rather than hiding it in the tolerance. I also used a gravity-free elastic reflection with timesteps aligned to impact. Absolute tolerance was 1e−11 and relative tolerance 1e−13, allowing roundoff while remaining well below the integration error.

Conservation properties

I checked momentum and kinetic energy in gravity-free flight, and kinetic energy across elastic wall collisions. Wall impacts do not conserve particle momentum. For inelastic impacts, I checked $\Delta K = -m(1-e^2)vn^2/2$. Absolute tolerance 1e−12 and relative tolerance 1e−13 covered the few arithmetic operations without allowing physical energy drift.

Convergence order

At T = 1.5 s, four successively halved timesteps gave position errors of about 0.45984, 0.22992, 0.11496, and 0.05748 m. I used the uncorrected error to calculate $q = \log_2[E(h)/E(h/2)]$, which was approximately one. Position and velocity tolerances were 1e−11; the order tolerance was 1e−9, based on roundoff propagation. This establishes convergence only for smooth, contact-free flight.

Symmetry and invariance

I compared mirrored trajectories using $x' = W - x$ and $vx' = -vx$, including elastic and inelastic contacts. I also compared motion with masses of 0.125 and 128 kg. The absolute tolerance of 5e−10 allowed paired roundoff propagation over 400 steps; contact identities and step counts were exact.

Limiting cases

I tested uniform motion and rest at zero gravity, e = 0 impacts with follow-up steps, and positive restitution approaching zero. Tolerances were 2e−11 m, 1e−13 m/s, and 1e−12 J. Small positive responses used relative-only tolerance of 1e−12 and positivity checks, so an incorrectly zeroed response could not pass.

Failures and deliberate defect

The five physics suites passed initially. During the function audit, a new report test failed because it used the wrong dictionary key. Correcting the key fixed it without changing expected values or tolerances. The complete baseline then passed 58 methods. Removing right-wall restitution caused 10 methods to fail, with 309 failure entries across subcases, as predicted. After byte-for-byte restoration, all 58 passed again.